

PRIOR — SHARED, UNCHANGED ACROSS ALL PHASES

Score $\mathcal{S} = \{(p_i, b_i, d_i)\}$
known in every phase

Score graph $\rightarrow L_G$
spectral kernel
 $K_G(s) = U g(\nu; s) U^\top$

Music model (Phase 0)
per-note embedding h_i

Feature kernels
 $\sum_f \text{diag}(c_f) \otimes X_f X_f^\top$
25 score features, h_i

Joint kernel
 $K = B \otimes K_G + \sum_f \text{diag}(c_f) \otimes X_f X_f^\top + \text{diag}(\varsigma^2) \otimes I$

Coregionalization B
couples the components of z

the same prior, now the structured component of $p(z)$

$p(z)$

Non-conjugate
Laplace / VI over z
score-based init; a
closed-form after

Posterior over
every note's z
+ amplitudes
posterior-predictive
resynthesis

Waveform $x \in \mathbb{R}^M$
no performance MIDI

LIKELIHOOD — THE ONLY PART THAT CHANGES

Audio likelihood

$x = \Phi(z) a + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, K_n)$
amplitudes a marginalised *exactly*:
 $p(x | z) = \mathcal{N}(\Phi \mu_a, \Phi \Sigma_a \Phi^\top + K_n)$

Phase 3 (not yet implemented; dashed). The observation is the waveform itself, so there are no per-note observations and no visible/hidden split: given only the score and the audio, every note's expression is latent and inferred jointly. Conjugacy is lost — the price of the audio likelihood.